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Ronan Tunney

https://ronantunney.ca

Waterloo Engineering grad with 4 years' experience delivering robust software applications and tools.

SKILLS & EXPERIENCE

Programming Languages	C#, SQL, R, Python, C/C++, JavaScript, Visual Basic, Bash
Frameworks	.NET, React, XML, JSON, REST, WPF, HTML/CSS
Tools	Git, Docker, Azure DevOps, AWS, Jira, Jenkins, Visual Studio
Competencies	Statistics, Numerical Modelling, Electronics, Mechanical Design

EXPERIENCE

Abbott Point of Care — Dec 2021 - Apr 2024
Intermediate Data Engineer — *Ottawa, ON*

- Developed a .NET-based wafer-map editor to replace legacy 32-bit tooling, eliminating VM dependencies and streamlining file editing across .map, .vik, and e142 formats. Empowered engineers with database integration, template overlays, edge exlusion tools, and undo/redo support.
- Architected a version-aware installer and update system using Squirrel, enabling seamless Dev/Val/Prod deployment and side-by-side installs - cutting update and deployment time for internal apps.
- Revamped the data engineering team's core .NET library used in Equipment Integration apps, reducing latency and improving UX through performance tuning, bug fixes, and custom WPF control rework.
- Integrated a Filmetrics F40 spectrometer with a C++ wafer inspection GUI by wrapping vendor DLLs in a C interface and running it in a separate thread for real-time measurements and calibration.

Abbott Point of Care — Sep 2020 - Dec 2021
Associate Engineer — *Ottawa, ON*

- Met stringent deadlines leading the development, validation and deployment of three R packages which formed the backbone of a generalized ETL pipeline. Earned promotion for strong performance.
- Generated thousands of finished goods and qualification reports for various blood testing cartridges using R and Microsoft SQL Server to develop a dynamic data processing and statistical analysis backend.
- Eliminated engineering time taken for the calibration of cardiac marker blood testing products by writing a multidimensional optimization algorithm in C++ which consistently delivered optimal results.

Ontario Die International — Apr 2019 – Aug 2019
Robotics Developer (Internship) — *Kitchener, ON*

- Enhanced the project team's debugging and data retention capabilities by proposing and then building a system to record and playback robot perception, sensor data and camera footage.
- Established novel machine learning data by writing a C++ process to aggregate steel forming data.
- Improved the onboarding process by developing and documenting a bash alias repository.
- Designed real world test cases which exposed flaws in new manufacturing software features.

Avidbots Corp. — Aug 2018 – Dec 2018
Software Developer (Internship) — *Kitchener, ON*

- Enhanced the reliability of company robots by writing a Python back-up system which would recover customer setting files in the event of a failed startup.
- Built a CI/CD pipeline which detected regressions in the company's cleaning plan tools using Jenkins, Docker and Bash scripts.
- Halved the size of robot Docker images by writing a smart dependency handler to separately install compilation and runtime packages.

EDUCATION

University of Waterloo *September 2015 - April 2020*
B.S. in Mechanical Engineering & Options in Management and Mechatronics
With Distinction | *Dean's Honours List* | *U Sports Academic All-Canadian*

ADDITIONAL PROJECTS

- **Tooner.io** Multipage React app providing guitar players with the tools to learn songs fast.
- **Interactive Travel Map** React app displaying geotagged photos on an interactive map and gallery.
- **Physics Engine** Programmed a rigid body physics engine with no external dependencies in JavaScript.
- **C++ Fluid Simulations** Simulated dynamic fluid behavior in C++ and OpenGL.
- **Robot Simulations** Set up a virtual robot simulation environement using Gazebo and ROS.